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HAPPINESS QUOTIENT

The start-up is the first in India focusing on Gallium Nitride (GaN) technology, contributing to the Centre's goal of an ecosystem in semiconductors

Between them, AGNIT's team of seven co-founders, which includes a semiconductor industry veteran, have expertise in GaN materials, manufacturing processes and radio frequency and power devices

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ments in IISc have progressed from manufacturing GaN material to making a device, integrating it in a system and testing it. Here, 'device' refers to electronic parts like switches that sit on the motherboard of the products we use.

Research at IISc began in 2005 and the first GaN project was sanctioned in 2009, following which a Metalorganic Chemical Vapor Deposition (MOCVD) reactor was set up. This is a process by which thin layers of the GaN compound are deposited on silicon wafers from which devices are fabricated in an intricate process using state-of-the-art equipment. For instance, the thickness of a human hair is about 100 microns; in comparison, the thickness of the blade that cuts the GaN-coated wafers into devices is 25 microns. "Those kinds of fabs are non-existent in India in the private sector," explains Raghavan.

Now, as the technology world is at a cusp in wireless communications, researchers at IISc are ready for a piece of the action. In 2019, a group of faculty members and students from two departments founded AGNIT Semiconductors. The market opportunity for GaN components globally is estimated to be $3 billion by 2026 and growing rapidly, points out Hareesh Chandrasekar, CEO and co-founder of AGNIT. "There is a strong RF design ecosystem in India. And a design ecosystem needs a fab," he adds. That's the gap the start-up—backed by IISc's Foundation for Science Innovation and Development and an angel investor—is trying to fill. For instance, it is looking to fabricate components for use in 5G telecommunications and wireless transmitters.

The focus at IISc is equally towards fostering a GaN ecosystem in India, given the semiconductor mission announced in December 2021 with an outlay of Rs 76,000 crore. "Very few countries are working on this," says Muralidharan Rangarajan, emeritus professor at IISc and former director of the Solid State Physics Laboratory at DRDO. "We are not replacing silicon everywhere. We are complementing it. We are not replacing your microprocessor in the laptop with GaN," he explains. "But one key component for power electronics is the switch. That will get replaced with gallium nitride."

Soon, FSID-IISc expects to commission a low-volume production fab for GaN at the IISc campus. Semiconductors are a long-gestation business but given the emerging market for EV chargers and 5G telecom applications, AGNIT's timing could be fab. ■



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